

Q1. Layers – Multi-Layer

Objective: Build a multi-layer map with regional shading layer for average delivery hours for South Indian states.

Simulated dataset

state	avg_delivery_hours	lat	lon
Tamil Nadu	28	11.1	78.6
Karnataka	34	15.3	75.7
Kerala	31	10.5	76.3
Andhra Pradesh	39	15.9	79.7
Telangana	36	17.4	79.0
Maharashtra	27	19.7	75.7
Gujarat	25	22.3	72.6

Code (folium, layered)

```
m = folium.Map(location=[16,78], zoom_start=10)
folium.Choropleth(geo_data=india, data=india_delivery_data,
                  columns=["state","avg_delivery_hours"],
                  fill_color="YlOrRd", legend_name="Avg Delivery Hours",
                  marker_layer = folium.FeatureGroup(name="Markers"))
for _, row in india_delivery_data.iterrows():
    folium.CircleMarker([row.latitude,row.longitude],
                        popup=f"{row.state}: {row.avg_delivery_hours}h")
marker_layer.add_to(m)
folium.LayerControl().add_to(m)
```

